

# Nathan Tsao

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## EDUCATION

### University of Texas at Austin

Aug 2023 – May 2025

*Masters of Science:* Engineering (Machine Learning & AI)

GPA: 3.74

*Thesis:* Physics-informed neural differential equations for electrical network modeling

### University of Illinois Urbana-Champaign

Aug 2019 – May 2022

*Bachelors of Science:* Mechanical Engineering

GPA: 3.87

## SKILLS

**Programming:** Python | TypeScript | SQL | C++

**ML/AI:** PyTorch | JAX | LangChain | Ray (RLlib)

**Infrastructure:** AWS | GCP | Docker | Databricks | FastAPI

**Languages:** English | Mandarin

## WORK EXPERIENCE

### TAO Digital Solutions

Mar 2026 – Current

Applied Scientist

Taipei, Taiwan

- Led development of two production sales teams AI workflows deployed on Databricks and Kubernetes using speech-to-text and language models for document field prediction, resulting in over \$100k in annual savings.
- Built AI platform leveraging LLMs for code generation to automate large-scale enterprise data ingestion and analysis workflows, reducing customer onboarding time from 90 days to 1 week.

### NASA Ames Research Center

June 2025 – Aug 2025

Autonomous Aircraft Operations Research Intern

Mountain View, CA

- Developed a multi-agent reinforcement learning framework (Ray RLlib) for autonomous aircraft routing, reducing average travel time across agents by 30% in simulation.

## SELECTED PROJECTS

### TranslateNets: Translation SaaS

Jan 2026

Personal Project

[translatenets.com](http://translatenets.com)

- Built multimodal translation platform supporting text, image, and document translation workflows across 60+ languages with OCR processing and translation APIs deployed on GCP CloudRun.
- Developed REST APIs and Python SDK enabling automated language detection and provider-agnostic LLM translation workflows with API key-based authentication and usage tracking.

### Causal Diffusion Guidance

May 2025

Statistical Machine Learning Final Project

- Developed causal diffusion framework for counterfactual image generation from structural causal models (SCMs) across arbitrary intervention settings.
- Trained distributed PyTorch diffusion models on two A100 GPUs and demonstrated causally consistent counterfactual generation on benchmark image datasets.

## PUBLICATIONS

Neary, Cyrus, **Nathan Tsao**, and Ufuk Topcu. *Neural port-hamiltonian differential algebraic equations for compositional learning of electrical networks*. 2025 IEEE 64th Conference on Decision and Control (CDC). IEEE, 2025.

Cooper, Geffen\*, **Nathan Tsao\***, Filippos Fotiadis, Ufuk Topcu, and Radu Marculescu. *Learning from Sparse and Asynchronous Data Streams for Batteryless Sensors*. Preprint.

## RESEARCH EXPERIENCE

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### **Autonomous Systems Group:** UT Austin

Dec 2023 – May 2025

Graduate Research Assistant

Austin, TX

- Designed and implemented novel physics-informed neural network architectures in JAX for digital twins, achieving  $>10\times$  accuracy improvements over baseline deep learning models.
- Developed a low-power human activity recognition model optimized for batteryless sensors using PyTorch, resulting in 15-50% relative improvement over baselines.

### **Hybrid Robotics Group:** UC Berkeley

May 2022 – Jan 2023

Visiting Research Assistant

Berkeley, CA

- Implemented a reinforcement learning locomotion balancing controller for tailed quadruped robots using PyTorch and IsaacGym.